Galaba Vamsi

Mechatronics at IIT Bhilai. Machine learning, computer vision, and applied AI.
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SUMMARY

Graduating May 2027. I build and evaluate ML systems: wearables recognition (Kaggle, 5th of 232 teams), a voice-clone detector (hackathon finalist), vision-language price prediction, and I/Q signal classification. 15 merged pull requests to open-source robot-data and evaluation tools; I wrote and maintain inspect-robots-wandb on PyPI. Co-author of 3 papers.

EDUCATION

B.Tech. in Mechatronics, Indian Institute of Technology Bhilai | **Coursework:** AI and ML, Deep Learning for Computer Vision, Digital Image Processing, Optimal Control, Advanced Control Theory, DSA, OOP. Aug 2023 – May 2027

PROJECTS

WEAR Dataset Challenge @HASCA 2026 (Kaggle) Wearable exercise recognition from one-second IMU and video tiles of four unseen subjects (macro-F1 over 19 classes): 5th of 232 teams on the final private leaderboard (0.93137; 7th public, 0.93825). A fork of a public notebook; my additions took my reproduction from 0.92497 to 0.93825 public: per-subject whitening, whitened video-continuity link features, ZCA video input, and a multi-run ensemble decode. A global whitening that raised out-of-fold score but lowered the leaderboard score was a leak, and I replaced it with per-subject whitening. Built with Claude. Leaderboard

Pehchaan (team of 2; top-30 finalist of 4,723, RAKSHAM (IIT Delhi × Amazon), 2026) UPI voice-clone fraud defence; I led ML and backend.

- Fine-tuned XLS-R-300M + AASIST on consented Telugu, Hindi, and English speech and clones from 13 generators: 4.8% EER vs 34.3% off the shelf (6.0% on 6 unseen generators).
- A 48-call real-carrier test exposed a gap (24.6% EER). Learning the speakerphone channel and retraining cut it to 6.8%, clones caught 48% to 75%, 0% of genuine callers flagged. In simulation the risk engine holds 73% of scams vs 0.21% of genuine payments.

Amazon ML Challenge (team; 2025) Product price prediction with a Qwen2-VL 7B and ViT/RoBERTa two-tower model; team rank 700 (20,000+ registered participants). Also AIR 200 in 2026.

Arista Wi-Fi RRM (team; Inter IIT Tech Meet 14.0, 5th place, Dec 2025) I/Q-based interference classification (STFT, DBSCAN, Dual-CNN; 16 classes) for client-aware radio resource management in NS-3 on a 5-AP, 50-client 802.11k/v topology. Slides

Lunar Crater Detection (2026) Reproduced Zhou et al. (2018) on lunar DEMs: F1 0.839 on one labelled region (parameters tuned on that region). GitHub

AntennaNet (2026) Inverse antenna design: surrogate network plus KD-tree anchoring over about 2,000 simulated S11 spectra, extended to 12 × 12-pixel layouts. GitHub

Electron-GNN (team, in progress, 2026) Two-tower GATv2 graph network predicting absorption-spectrum peaks from molecular geometry; so far trained and checked on two small molecules (water and ammonia). GitHub

EXPERIENCE

Research Intern IIT Bhilai | Advisor: Dr. Arzad Alam Kherani | RIS-SIM v2 | v1 May 2025 – Jul 2025

- Built RIS-SIM, an open-source Python emulator for RIS-assisted radio links; designed RIS unit cells in CST Studio Suite (IEEE INDICON 2025, co-author). Rebuilt it in 2026 as RIS-SIM v2 (103 tests), then as sremu, an IQ-level emulator with about 2,100 tests (GitHub).

Co-founder, Human Slop Anti-AI social app | Web app | Play Store 2026 – Present

- Co-building an Expo app (web and Android), a Fastify API, and PostgreSQL on Supabase; live on Google Play (50+ downloads). Added a keystroke-timing score and a device biometric check; both are heuristics, not proof of authorship.

OPEN SOURCE

Inspect Robots and inspect-robots-wandb Robocurve (YC S26) | Repository | PyPI Sep 2026 – Oct 2026

- Evaluation framework for physical-AI policies. 2 merged PRs (a configurable LLM retry policy across 5 wire protocols; git state no longer logged as clean when unknown), 3 open. Wrote and maintain inspect-robots-wandb, a W&B plugin on PyPI with a 100% coverage gate and trusted publishing.

HFlow Hebbian Robotics (YC S26) | Repository Sep 2026 – Oct 2026

- 8 merged PRs to an SDK that verifies robot training data: conversion keeps data whole or refuses it with a reason, e.g. depth videos that would silently drop from 12-bit to 10-bit (#400). A benchmark PR reported no safe speedup and changed nothing.

PUBLICATIONS

- **Co-author** — *Inspect Robots: Evaluating the Capabilities and Safety of Embodied AI*, arXiv:2610.06306, 2026; accepted to the SPAIS Workshop at CoRL 2026 (poster). arXiv
- **Co-author** — *Experience with RF Energy Harvesting-Driven Self-Powered RIS*, IEEE INDICON 2025.
- **Co-author** — *A systematic literature review on simulation models and deployments for reconfigurable intelligent surfaces*, Ad Hoc Networks (Elsevier), 2026.
- **Sole author (preprint draft, arXiv submission pending)** — *Timed RIS Control in the Loop: A Sample-Accurate IQ-Level Emulator with Characterized Timing*, 2026. Draft

SKILLS

ML and data: PyTorch, PyTorch Geometric, LightGBM, OpenCV, TFLite, NumPy, CuPy, Apache Arrow, Weights & Biases, scikit-image, Qwen2-VL, ViT/RoBERTa, XLS-R/AASIST
Languages: Python, SQL, TypeScript/JavaScript, C/C++, Rust (open-source contributions), GLSL
Tooling: FastAPI, PostgreSQL, Docker, AWS, Git, Linux, GitHub Actions, pytest, Ruff, mypy, PyPI trusted publishing
Scientific computing and simulation: NS-3, MEEP (FDTD), CST Studio Suite, GNU Radio, MATLAB/Simulink, MuJoCo

ACHIEVEMENTS

5th of 232 teams, WEAR Dataset Challenge @HASCA 2026 (Kaggle, final private leaderboard, macro-F1 0.93137) | **AIR 200** (2026) and **AIR 700** (2025; 20,000+ participants), Amazon ML Challenge (team) | **5th place**, Arista Networks Wi-Fi challenge, Inter IIT Tech Meet 14.0 (team, Dec 2025) | **Top 1%**, JEE Advanced | **PMSS** scholarship (2024) | **Subroto Mukherjee** Scholarship (IAFBA)